Aerobic Phenylacetate Catabolism: The Phenylacetyl-CoA Pathway

A review-style synthesis for a molecular-biology audience

1. Executive summary

Aerobic phenylacetate (PA) catabolism is a widespread bacterial strategy for funnelling a large family of aromatic substrates — phenylalanine, 2-phenylethylamine, styrene, ethylbenzene, phenylacetyl-esters, lignin- and plastic-derived aromatics — into central metabolism. Its defining and counter-intuitive feature is that the aromatic ring is **not** attacked while it is a free acid. Instead, phenylacetate is first activated to the coenzyme-A thioester **phenylacetyl-CoA**, and every downstream intermediate remains a CoA thioester. The CoA-bound ring is then **epoxidized** (not dihydroxylated) at the 1,2 position by a multicomponent, ferritin-like diiron oxygenase (PaaABC(D)E), isomerized to a seven-membered oxygen heterocycle (**oxepin-CoA**) by the crotonase-family enzyme PaaG, **hydrolytically opened and the resulting semialdehyde oxidized** by the bifunctional enzyme PaaZ, and finally degraded through a **β-oxidation-like** sequence (PaaF → PaaH → PaaJ) to two central-metabolic products: **acetyl-CoA and succinyl-CoA** (Teufel et al. 2010, P 20660314; Ferrández et al. 1998, P 9748275).

This "hybrid" route — aromatic activation as a thioester followed by epoxide/oxepin chemistry and β-oxidation — is mechanistically distinct from the classical ring-dihydroxylation (dihydrodiol) paradigm of aerobic aromatic degradation, and it operates in roughly one-sixth of all sequenced bacterial genomes, including *Escherichia coli*, *Pseudomonas putida*, and numerous pathogens where it is increasingly linked to virulence and antimicrobial tolerance.

2. Definition and biological boundaries

2.1 What is included

The core system, as scoped here, is the **reusable phenylacetyl-CoA reaction chain** comprising seven catalytic stages:

1. **Activation** — phenylacetate → phenylacetyl-CoA (PaaK, an ATP-dependent, adenylate-forming ligase).
2. **Ring 1,2-epoxidation** — phenylacetyl-CoA → ring-1,2-epoxyphenylacetyl-CoA (PaaABC(D)E multicomponent oxygenase).
3. **Epoxide isomerization** — epoxide → oxepin-CoA (PaaG).
4. **Hydrolytic ring opening** + **semialdehyde oxidation** — oxepin-CoA → 3-oxo-5,6-dehydrosuberyl-CoA semialdehyde → 3-oxo-5,6-dehydrosuberyl-CoA (bifunctional PaaZ).
5. **Enoyl-CoA hydration** — (after one thiolytic shortening) 2,3-dehydroadipyl-CoA → 3-hydroxyadipyl-CoA (PaaF).
6. **3-hydroxyacyl-CoA oxidation** — 3-hydroxyadipyl-CoA → 3-oxoadipyl-CoA (PaaH, NAD^+).
7. **Thiolytic cleavage** — 3-oxoadipyl-CoA → succinyl-CoA + acetyl-CoA (PaaJ thiolase).

2.2 What lies at, but outside, the boundary

- **Upstream "peripheral" pathways** that *generate* phenylacetate — e.g. the Pea/Ped routes that convert 2-phenylethylamine and 2-phenylethanol to phenylacetate in *P. putida* (Arias et al. 2008, P 18177365), styrene side-chain oxidation, and phenylalanine transamination/decarboxylation — feed into, but are not part of, the core catabolon. They are *convergent* funnels; the phenylacetyl-CoA chain is the shared *core*.
- **Regulation and transport** — the PaaX repressor, the PaaY accessory protein, and PA/PA-CoA transporters govern *when and whether* the pathway runs but do not carry out ring chemistry (Ferrández et al. 1998, P 9748275). They are accessory.

- **Detoxification/stress functions** — the reactive early intermediates (epoxide, oxepin, ring-opened semialdehyde) impose a chemical burden; proteins that buffer this (e.g. PaaD, PaaY assignments vary) are protective rather than catalytic to the main flux.
- **Anaerobic aromatic catabolism** (benzoyl-CoA reductase pathways) is a *separate* system that is frequently confused with this one because both use CoA thioesters and a "benzoyl/phenylacetyl-CoA" logic — but the aerobic PA pathway is strictly O_2-dependent (the oxygenase step), whereas the anaerobic route reductively dearomatizes the ring using ATP and low-potential electrons. They should be treated as convergent-but-independent solutions.
- **Mammalian/eukaryotic "phenylalanine catabolism" is a different pathway entirely.** In humans, phenylalanine is hydroxylated to tyrosine and degraded through 4-hydroxyphenylpyruvate → **homogentisate** → maleylacetoacetate → fumarate + acetoacetate; defects in this route cause phenylketonuria and alkaptonuria, and nitisinone therapy exploits it (Ranganath et al. 2022, P 36167967; Chen et al. 2024, P 39258552). This is mechanistically unrelated to the bacterial phenylacetyl-CoA epoxide/oxepin pathway and should never be conflated with it. The bacterial paa system, by contrast, receives phenylalanine only after it has been routed to **phenylacetate** (e.g. via transamination/decarboxylation or the phenylethylamine route), not via tyrosine.

2.3 Competing definitions

Two framings coexist in the literature: (i) the **"phenylacetyl-CoA catabolon"** framing (Luengo/García school), which emphasizes the whole convergent network of peripheral routes plus the central core; and (ii) the **"paa pathway / hybrid pathway"** framing (Díaz/Fuchs school), which emphasizes the biochemically defined core reaction chain (Ferrández et al. 1998, P 9748275; Teufel et al. 2010, P 20660314). This review adopts the second, narrower definition for the core while naming the convergent inputs.

3. Mechanistic overview

3.1 The obligatory reaction sequence (best current model)

The Teufel et al. (2010) reconstitution remains the definitive mechanistic model (P 20660314). In order:

Step 1 — Activation (PaaK). Phenylacetate + ATP + CoA → phenylacetyl-CoA + AMP + PP_i. This is the committed, obligatory entry step; without it no ring chemistry occurs, and phenylacetyl-CoA is also the physiological inducer of the operon (see §4.6).

Step 2 — Ring 1,2-epoxidation (PaaABC(D)E). Molecular O_2 and NADPH are used to install a **1,2-epoxide** across the aromatic ring, giving ring-1,2-epoxyphenylacetyl-CoA. Thioesterification to CoA is what makes the otherwise inert benzene ring susceptible to epoxidation; the CoA carbonyl withdraws electron density and stabilizes the non-aromatic product. This is the single **O_2-requiring, obligatory** step and the reason the whole pathway is "aerobic."

Step 3 — Isomerization to oxepin-CoA (PaaG). The strained, reactive epoxide is ring-expanded/isomerized to **2-oxepin-2(3H)-ylideneacetyl-CoA (oxepin-CoA)**, a seven-membered O-heterocyclic enol ether. The epoxide ⇌ oxepin equilibrium is a distinctive chemical signature of this pathway.

Step 4 — Hydrolytic ring opening and semialdehyde oxidation (PaaZ, bifunctional). PaaZ hydrolytically opens oxepin-CoA to a ring-opened **3-oxo-5,6-dehydrosuberyl-CoA semialdehyde**, then its aldehyde-dehydrogenase domain oxidizes the semialdehyde (using $NADP^+$) to **3-oxo-5,6-dehydrosuberyl-CoA**. Cryo-EM shows PaaZ transfers the unstable intermediate between its two active sites by electrostatic channeling (Sathyanarayanan et al. 2019, P 31511507).

Steps 5–7 — β-oxidation to central metabolites. The C8 dicarboxylic 3-oxo-5,6-dehydrosuberyl-CoA undergoes a first thiolytic cleavage (releasing acetyl-CoA) to a C6 2,3-dehydroadipyl-CoA, which is then hydrated by **PaaF** to 3-hydroxyadipyl-CoA, oxidized by

PaaH (NAD^+) to 3-oxoadipyl-CoA, and cleaved by the **PaaJ** thiolase to **succinyl-CoA** + **acetyl-CoA**. In *E. coli* the single thiolase PaaJ performs both thiolytic cleavages; PaaG can also participate in the open-chain isomerizations.

3.2 Obligatory vs. conditional vs. accessory

- **Obligatory:** PaaK activation; PaaA/PaaC oxygenase + PaaE reductase (epoxidation); PaaZ ring opening/oxidation; the PaaF/PaaH/PaaJ β-oxidation module. Deleting any of these blocks growth on PA.
- **Conditional / context-dependent:** the specific peripheral input route (styrene, phenylethylamine, phenylalanine, etc.); PaaG's downstream isomerase roles; the identity of the thiolase(s) when paralogs exist.
- **Accessory:** PaaX repression, PaaY, transporters, and putative chaperone/protective factors (PaaD assignments remain uncertain).

---

4. Major molecular players and active assemblies

4.1 PaaK — phenylacetate-CoA ligase (activation)

A member of the ATP-dependent **adenylate-forming enzyme (AFE) superfamily**. Structures of *Burkholderia cenocepacia* PaaK1/PaaK2 captured ATP- and adenylate-intermediate complexes and revealed an N-terminal microdomain proposed to recruit downstream Paa enzymes, plus a substrate pocket that tunes aryl-substituent tolerance (Law & Boulanger 2011, P 21388965).
Some organisms carry paralogous ligases (paaK1/paaK2) with divergent kinetics and substrate scope, while all downstream enzymes are single-copy.

4.2 PaaABC(D)E — multicomponent phenylacetyl-CoA 1,2-epoxidase (ring activation)

A distinct multicomponent oxygenase (Ferrández et al. 1998, P 9748275 ; Teufel et al. 2010,

).

P 20660314

Functional assignments from biochemistry and structural work: - **PaaA** — catalytic oxygenase subunit bearing a **ferritin-like (four-helix bundle) carboxylate-bridged diiron center** that activates O_2 and epoxidizes the ring. - **PaaC** — structural/partner subunit forming the (PaaA·PaaC) heterodimeric core, ferritin-like fold but catalytically inactive. - **PaaB** — small accessory subunit required for full activity. - **PaaE** — the **reductase**: an NAD(P)H-, FAD- and [2Fe-2S]-containing ferredoxin-reductase that supplies electrons to the diiron site. - **PaaD** — role least certain; implicated in complex assembly / iron–sulfur handling or protection rather than catalysis. This assembly is the mechanistic heart of the pathway and its O_2 dependence.

4.3 PaaG — 1,2-epoxyphenylacetyl-CoA isomerase (oxepin formation)

A **crotonase (enoyl-CoA hydratase/isomerase) superfamily** enzyme that interconverts the epoxide and oxepin-CoA and can also catalyze downstream open-chain isomerizations.

4.4 PaaZ — bifunctional oxepin-CoA hydrolase / semialdehyde dehydrogenase (ring opening)

A fusion of a **MaoC-like (R)-hydratase/enoyl-CoA-hydratase-type domain** (ring-opening "hydrolase" activity) and an **aldehyde-dehydrogenase (ALDH) domain** ($NADP^+$-dependent semialdehyde oxidation). Cryo-EM revealed a trilobed assembly of three domain-swapped dimers and an **electrostatic-pivoting channeling mechanism** that shuttles the reactive ring-opened intermediate between the two active sites (Sathyanarayanan et al. 2019,

).

P 31511507

Fusion + channeling is the structural solution to the instability/toxicity of the ring-opened aldehyde.

4.5 PaaF / PaaH / PaaJ — the β-oxidation module (mineralization)

- **PaaF** — crotonase-family **enoyl-CoA hydratase** (produces 3-hydroxyadipyl-CoA).
- **PaaH — (3S)-3-hydroxyacyl-CoA dehydrogenase (NAD^+)** (→ 3-oxoadipyl-CoA).
- **PaaJ — 3-oxoadipyl-CoA thiolase** performing thiolytic C–C cleavage(s) that release acetyl-CoA and ultimately yield succinyl-CoA. These are homologous to fatty-acid β-oxidation enzymes (Ferrández et al. 1998,

9748275),
reflecting recruitment of a pre-existing enzymatic toolkit.

4.6 Regulatory/accessory players (outside the core chain)

- **PaaX** — repressor of the catabolic promoter; **phenylacetyl-CoA is the true inducer** (relieves PaaX repression), shown in *E. coli* (Ferrández et al. 1998, 9748275) and independently in *P. putida* U (García et al. 2000, 11010921).
- **PaaY**, transporters, and PA-generating peripheral enzymes complete the physiological context.
- **Carbon catabolite repression** overlays the PaaX/PA-CoA logic: in *P. putida* KT2440 the *paaA* promoter is strongly PA-inducible but repressed by glucose (though not by succinate or pyruvate), so preferred carbon sources dominate over aromatic catabolism (Kim et al. 2007, 18156794). This global regulation sits outside the core reaction chain but governs when the pathway is deployed.

---

5. Evolutionary and cell-biological variation

5.1 Phylogenetic distribution

The pathway is broadly distributed across Proteobacteria (α, β, γ), Actinobacteria, and others, present in ~16% of sequenced bacterial genomes at the time of the Teufel study (20660314).
The **core enzyme set (PaaK, PaaABCE, PaaG, PaaZ, PaaFHJ) is highly conserved**, consistent with a single ancestral solution reused across lineages. Comparative genomics reinforces this: in a survey of 80 Burkholderiales genomes, the **phenylacetyl-CoA ring-cleavage pathway was among the most widespread central pathways (≥60% of genomes)**, on par with protocatechuate/catechol ortho-cleavage and homogentisate cleavage (Pérez-Pantoja et al.

2012,

P 22026719

).

Notably, catabolic clusters showed **genus-specific positional ordering of genes — a signature of recent evolutionary rearrangement — and a bias toward secondary chromosomes (chromids)**, and environmental isolates carried more catabolic genes than host-associated strains. Thus the *catalytic core is ancient and conserved* while its *genomic packaging (operon order, chromosomal location) is evolutionarily labile* and lifestyle-shaped.

5.1b Deep origin and which family members are most representative

The pathway is a **mosaic of recruited enzyme families** rather than a single novel invention: PaaK belongs to the ancient ATP-dependent adenylate-forming (acyl-CoA ligase/luciferase) superfamily; PaaG and PaaF belong to the crotonase (enoyl-CoA hydratase/isomerase) superfamily; PaaH is a classic 3-hydroxyacyl-CoA dehydrogenase and PaaJ a thiolase — the last three being direct borrowings from **fatty-acid β-oxidation** (Ferrández et al. 1998,

P 9748275

).

The genuinely **derived, pathway-defining innovations** are the ferritin-like diiron **epoxidase (PaaA)** and the fused ring-opening/oxidizing enzyme **PaaZ**, whose oxepin-CoA chemistry has no close analog in central metabolism. For understanding the *ancestral* role of the expanded ligase family, the single-copy PaaK found in organisms with one ligase (e.g. *E. coli*) is the better representative of the core function, whereas paralog pairs (PaaK1/PaaK2 in *Burkholderia*) represent later, substrate-range-driven duplications (Law & Boulanger 2011,

P 21388965

).

5.2 Lineage-specific elaborations and replacements

- **Ligase duplication:** some organisms (e.g. *B. cenocepacia*) carry two ligases (PaaK1/PaaK2) with distinct kinetics/substrate range, while downstream steps stay single-copy (Law & Boulanger 2011,

  P 21388965

)

  — a classic pattern of expansion at the recruitment/entry node where substrate breadth is selected.
- **Cluster architecture varies:** gene order, operon splitting (e.g. *paaZ* separate from *paaABCDEFGHIJK* in *E. coli*), and regulator identity differ between taxa (Ferrández et al.

1998,

P 9748275).

- **Whole-cluster duplication and redundancy:** *Pseudomonas* sp. Y2 carries **two independent, functionally redundant paa clusters (paa1 and paa2)**; either alone supports growth on phenylacetate, and only the double deletion abolishes it. Their sequence divergence points to **independent evolutionary histories** (paa2's organization resembles *P. putida* KT2440 more than the co-resident paa1), likely tied to the adjacent styrene (*sty*) genes (Bartolomé-Martín et al. 2004,

  P 15474299).

  This illustrates the pathway as a duplicable, horizontally mobile cassette.
- **Continuity with dicarboxylate metabolism:** the lower, β-oxidation half of the pathway generates **adipyl/suberyl-CoA-type dicarboxylyl-CoA intermediates** that are chemically continuous with generic medium-chain dicarboxylic-acid catabolism. In *P. putida* KT2440, laboratory-evolved strains recruited the **paa cluster to metabolize adipate** (a nylon/polyester monomer), and its regulatory context overlaps with redundant β-oxidation genes (Ackermann et al. 2021,

  P 33965615)

  — underscoring both the modularity of the lower pathway and its biotechnological value for plastic-monomer upcycling.
- **Repurposing to new substrates:** a *Micrococcus luteus* isolate carries an unusual aromatic-degradation cluster containing *paa* genes (including a PaaK that also activates 1-naphthoic acid) proposed to degrade 1-methylnaphthalene via a **benzene-oxide/oxepin ring-opening** analogous to the PA route — a striking example of the epoxide/oxepin chemistry being recruited for polyaromatic catabolism (Iriani et al. 2025,

  P 40578825).
- **Biotechnological offshoot:** in *P. putida* the β-oxidation intermediates (3-hydroxy-phenylalkanoyl-CoAs) can be diverted into biodegradable aromatic polyhydroxyalkanoate plastics via the *pha* locus (García et al. 1999,

  P 10506180),

  showing how the core chemistry is co-opted.

5.3 Physiological-state variation and pathogenesis

Because bacteria are unicellular, "cell-type/tissue" variation is replaced by **physiological-state and niche** variation: - In pathogens, the pathway is induced under host-relevant conditions. In carbapenem-resistant *Acinetobacter baumannii*, exposure to human urine differentially regulates PA catabolism among ~112 shared adaptive genes (Escalante et al. 2024, P 39160175).
- Active PA catabolism in *A. baumannii* increases biofilm formation, surface adherence, efflux-pump activity, desiccation tolerance, and antibiotic MICs — linking a central carbon pathway to **virulence and drug tolerance** (Bhardwaj et al. 2025, P 40289051).
- PA-metabolic genes are associated with infection phenotypes across strains (Ross et al. 2024, P 39315786).
These illustrate that the same conserved core is deployed with different regulatory and phenotypic consequences depending on environment and lineage.

---

6. Constraints, dependencies, and failure modes

6.1 Mandatory ordering

The chemistry enforces a strict order: **activation → epoxidation → isomerization → ring opening → oxidation → hydration → oxidation → thiolysis.** Key constraints: - **CoA activation must precede ring attack.** The oxygenase acts on phenylacetyl-**CoA**, not free PA; the thioester is what enables epoxidation. This rules out a "free-acid dihydroxylation" route through this system. - **Epoxidation requires O_2** and reducing equivalents via PaaE. This is the point of strict aerobiosis and the mechanistic line separating this system from anaerobic benzoyl-CoA reduction. - **Isomerization before hydrolysis.** PaaZ acts on oxepin-CoA; the epoxide must first be isomerized by PaaG. - **Within PaaZ, hydrolysis precedes oxidation**, and the two are physically coupled by channeling (Sathyanarayanan et al. 2019, P 31511507)

— the semialdehyde is not released freely. - **β-oxidation order** (hydratase → dehydrogenase → thiolase) follows canonical fatty-acid logic; the 3-oxo group must be generated before thiolytic cleavage.

6.2 Reactive-intermediate burden (failure modes)

The epoxide, oxepin, and ring-opened semialdehyde are chemically reactive and potentially cytotoxic/genotoxic. Failure modes include: - **Bottlenecks that accumulate the epoxide/oxepin** (e.g. loss of PaaG or PaaZ) can be toxic and are proposed to contribute to virulence phenotypes when the pathway is partially active (Teufel et al. 2010, P 20660314). - **Channeling loss** (PaaZ mutants disrupting the electrostatic pathway) impairs growth, showing the intermediate cannot be safely handled in bulk solution (Sathyanarayanan et al. 2019, P 31511507). - **Uncoupled oxygenase** (PaaE without proper partners) risks futile O_2/NAD(P)H consumption and oxidative stress.

6.3 Evidence ruling out alternative paths

- The reconstitution of authentic epoxide and oxepin intermediates (rather than a catechol/dihydrodiol) rules out a classical dioxygenase route through this cluster (Teufel et al. 2010, P 20660314).
- Genetic induction studies (PA vs. PA-CoA) rule out free phenylacetate as the inducer and thereby argue the committed intermediate gates the pathway (García et al. 2000, P 11010921).

---

7. Controversies and open questions

1. **Exact subunit roles and stoichiometry of PaaABC(D)E.** The diiron-oxygenase assignment for PaaA and the reductase role of PaaE are well supported, but the precise function of **PaaD** (assembly? Fe–S maturation? protection?) and the working stoichiometry of the intact epoxidase complex remain incompletely resolved.

2. **Epoxide ⇌ oxepin equilibrium and PaaG's full catalytic repertoire.** How many of the isomerization steps PaaG performs in vivo, and how the equilibrium is biased toward productive flux, is still debated.
3. **Thiolase multiplicity and open-chain step assignments.** Whether one thiolase (PaaJ) performs both thiolytic cleavages in all organisms, and the exact order of open-chain hydration/isomerization steps, varies between reconstructions and organisms.
4. **Organism transferability.** Much detailed biochemistry comes from *E. coli*, *P. putida*, *Klebsiella*, and *Burkholderia*; extrapolating quantitative kinetics or regulatory logic across all paa-bearing taxa risks overgeneralization, especially where ligase paralogs or divergent regulators exist (Law & Boulanger 2011, P 21388965).
5. **Role in pathogenesis — correlation vs. causation.** Associations between PA catabolism and virulence/drug tolerance are increasingly reported (P 40289051; 39160175; 39315786), but whether the pathway *causes* these phenotypes or is co-regulated with them requires more direct, mechanistic tests.
6. **Substrate-range plasticity.** The *M. luteus* naphthalene-oxide/oxepin proposal (P 40578825) suggests the epoxide/oxepin chemistry generalizes beyond phenylacetate; how far the enzyme set can be re-tasked is an open evolutionary and biotechnological question.

---

8. Key references

- Teufel R, Mascaraque V, Ismail W, Voss M, Perera J, Eisenreich W, Haehnel W, Fuchs G. *Bacterial phenylalanine and phenylacetate catabolic pathway revealed.* PNAS. 2010. P 20660314 — definitive biochemical reconstitution; CoA-thioester/epoxide/oxepin model.
- Ferrández A, Miñambres B, García B, Olivera ER, Luengo JM, García JL, Díaz E. *Catabolism of phenylacetic acid in Escherichia coli. Characterization of a new aerobic hybrid pathway.* J Biol Chem. 1998.

P 9748275

— paa cluster, gene assignments, PaaX regulation, PA-CoA inducer.

- Sathyanarayanan N, et al. *Molecular basis for metabolite channeling in a ring opening enzyme of the phenylacetate degradation pathway.* Nat Commun. 2019.

P 31511507

— PaaZ cryo-EM structure and channeling mechanism.

- Law A, Boulanger MJ. *Defining a structural and kinetic rationale for paralogous copies of phenylacetate-CoA ligases from … Burkholderia cenocepacia J2315.* J Biol Chem. 2011.

P 21388965

— PaaK structures, AFE superfamily, ligase paralogy.

- García B, Olivera ER, Miñambres B, Carnicero D, Muñiz D, Naharro G, Luengo JM. *Phenylacetyl-coenzyme A is the true inducer of the phenylacetic acid catabolism pathway in Pseudomonas putida U.* FEMS/appl. 2000.

P 11010921

— inducer identity.

- Arias S, Olivera ER, Arcos M, Naharro G, Luengo JM. *Genetic analyses … conversion of 2-phenylethylamine and 2-phenylethanol into phenylacetic acid in Pseudomonas putida U.*
-

P 18177365

— peripheral (Pea/Ped) input routes.

- García B, et al. *Novel biodegradable aromatic plastics from a bacterial source … phenylacetyl-CoA catabolon.* J Biol Chem. 1999.

P 10506180

— catabolon concept / PHA offshoot.

- Iriani D, Allison J, Bugg TDH. *Degradation of brown coal and 1-methylnaphthalene by a Micrococcus luteus isolate … novel gene cluster containing paa genes.* 2025.

P 40578825

— benzene-oxide/oxepin chemistry re-tasked for polyaromatics.

- Bhardwaj et al. *Phenylacetic acid catabolism modulates virulence factors and drug resistance in Acinetobacter baumannii.* 2025.

  P 40289051

- Escalante et al. *Carbapenem-resistant Acinetobacter baumannii: metabolic adaptation and transcriptional response to human urine.* 2024.

  P 39160175

- Ross et al. *Phenylacetic acid metabolic genes are associated with … infection.* 2024.

  P 39315786

- Pérez-Pantoja D, Donoso R, Agulló L, Córdova M, Seeger M, Pieper DH, González B. *Genomic analysis of the potential for aromatic compounds biodegradation in Burkholderiales.* Environ Microbiol. 2012.

  P 22026719

  — comparative distribution of the phenylacetyl-CoA ring-cleavage pathway and cluster evolution.

- Ranganath et al. *Determinants of tyrosinaemia during nitisinone therapy in alkaptonuria.* 2022.

  P 36167967

  — boundary reference: the mammalian tyrosine/homogentisate route for phenylalanine, distinct from the bacterial paa pathway.

- Bartolomé-Martín D, et al. *Characterization of a second functional gene cluster for the catabolism of phenylacetic acid in Pseudomonas sp. strain Y2.* Gene. 2004.

  P 15474299

  — dual redundant paa clusters and independent evolutionary histories.

- Ackermann YS, et al. *Engineering adipic acid metabolism in Pseudomonas putida.* Metab Eng. 2021.

  P 33965615

  — functional overlap of the paa lower pathway with dicarboxylate metabolism; biotechnological co-option.

- Kim Y, Jeon Y, Park H, et al. *A green fluorescent protein-based whole-cell bioreporter for the detection of phenylacetic acid.* 2007.

P 18156794
— PA-inducible paaA promoter and glucose catabolite repression in P. putida KT2440.

Uncertainty flags: Subunit-level assignments within PaaABC(D)E (especially PaaD) and the precise open-chain step order derive from a limited set of model organisms; quantitative and regulatory details should not be assumed universal across all paa-bearing bacteria. Structural/mechanistic claims for PaaK, PaaZ, and the overall pathway are strongly supported; virulence links are so far largely correlative.